Log Analysis Lab Report

Linux System & Security Log Investigation Exercise

LEVEL 1 – BASIC (COMPULSORY)

Task 1: Command Explanations

- **ls**: Lists all files and directories in the current folder.
- **pwd**: Displays the absolute working directory path.
- **cd**: Navigates across file system directories.
- **cat**: Prints and concatenates file contents to standard output.

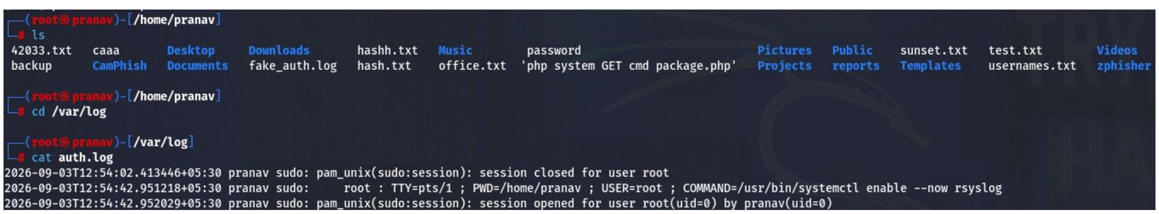
Task 2: Directory Exploration (/var/log)

Commands used:

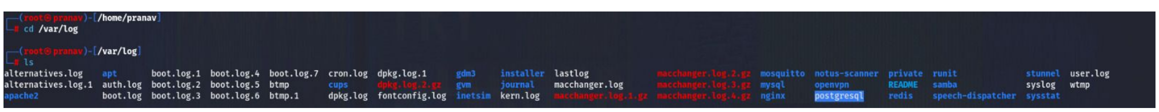
```
cd /var/log → ls
```

Key files found:
auth.log, syslog, kern.log, dpkg.log, alternatives.log, boot.log, apache2, mysql, postgresql

Screenshot: ls of /home/pranav before navigating to /var/log



Screenshot: cd /var/log → ls output



Task 3: File Creation

Commands used:

```
echo "testing day 3 work" > test.txt → cat test.txt
```

Output verified:

```
testing day 3 work
```

Screenshot: cat test.txt output

LEVEL 2 – INTERMEDIATE

Task 4: Log Reading (/var/log/syslog)

Log type:
Operating system level operational logs tracking service statuses, kernel events, and systemd daemons.

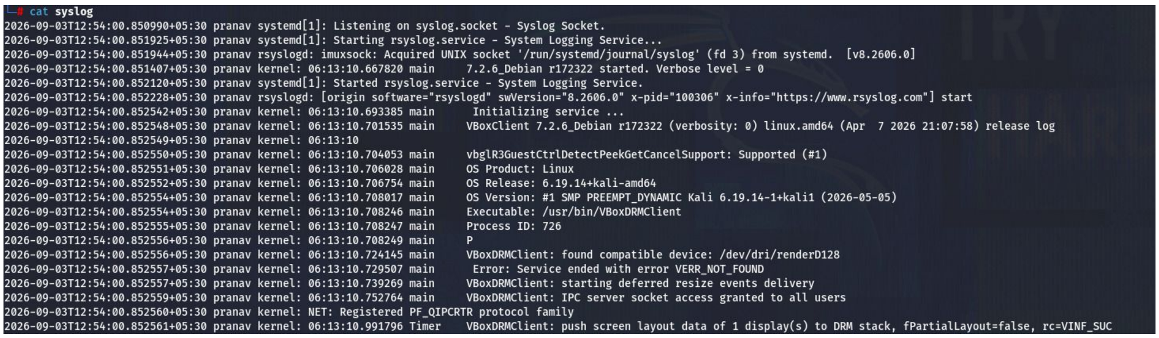
3 log entries:

```
systemd[1]: Starting rsyslog.service - System Logging Service...

rsyslogd: [origin software="rsyslogd" swVersion="8.2606.0"] start

kernel: OS Release: 6.19.14+kali-amd64
```

Screenshot: cat syslog output


```
└─# cat syslog
2026-09-03T12:54:00.850990+05:30 pranav systemd[1]: Listening on syslog.socket - Syslog Socket.
2026-09-03T12:54:00.851925+05:30 pranav systemd[1]: Starting rsyslog.service - System Logging Service...
2026-09-03T12:54:00.851944+05:30 pranav rsyslogd: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd.  [v8.2606.0]
2026-09-03T12:54:00.851407+05:30 pranav kernel: 06:13:10.667820 main     7.2.6_Debian r172322 started. Verbose level = 0
2026-09-03T12:54:00.852120+05:30 pranav systemd[1]: Started rsyslog.service - System Logging Service.
2026-09-03T12:54:00.852228+05:30 pranav rsyslogd: [origin software="rsyslogd" swVersion="8.2606.0" x-pid="100306" x-info="https://www.rsyslog.com"] start
2026-09-03T12:54:00.852542+05:30 pranav kernel: 06:13:10.693385 main      Initializing service ...
2026-09-03T12:54:00.852548+05:30 pranav kernel: 06:13:10.701535 main     VBoxClient 7.2.6_Debian r172322 (verbosity: 0) linux.amd64 (Apr  7 2026 21:07:58) release log
2026-09-03T12:54:00.852549+05:30 pranav kernel: 06:13:10
2026-09-03T12:54:00.852550+05:30 pranav kernel: 06:13:10.704053 main     vbglR3GuestCtrlDetectPeekGetCancelSupport: Supported (#1)
2026-09-03T12:54:00.852551+05:30 pranav kernel: 06:13:10.706028 main     OS Product: Linux
2026-09-03T12:54:00.852552+05:30 pranav kernel: 06:13:10.706754 main     OS Release: 6.19.14+kali-amd64
2026-09-03T12:54:00.852554+05:30 pranav kernel: 06:13:10.708017 main     OS Version: #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05)
2026-09-03T12:54:00.852554+05:30 pranav kernel: 06:13:10.708246 main     Executable: /usr/bin/VBoxDRMClient
2026-09-03T12:54:00.852555+05:30 pranav kernel: 06:13:10.708247 main     Process ID: 726
2026-09-03T12:54:00.852556+05:30 pranav kernel: 06:13:10.708249 main     P
2026-09-03T12:54:00.852556+05:30 pranav kernel: 06:13:10.724145 main     VBoxDRMClient: found compatible device: /dev/dri/renderD128
2026-09-03T12:54:00.852557+05:30 pranav kernel: 06:13:10.729507 main      Error: Service ended with error VERR_NOT_FOUND
2026-09-03T12:54:00.852557+05:30 pranav kernel: 06:13:10.739269 main     VBoxDRMClient: starting deferred resize events delivery
2026-09-03T12:54:00.852559+05:30 pranav kernel: 06:13:10.752764 main     VBoxDRMClient: IPC server socket access granted to all users
2026-09-03T12:54:00.852560+05:30 pranav kernel: NET: Registered PF_QIPCRTR protocol family
2026-09-03T12:54:00.852561+05:30 pranav kernel: 06:13:10.991796 Timer    VBoxDRMClient: push screen layout data of 1 display(s) to DRM stack, fPartialLayout=false, rc=VINF_SUC
```

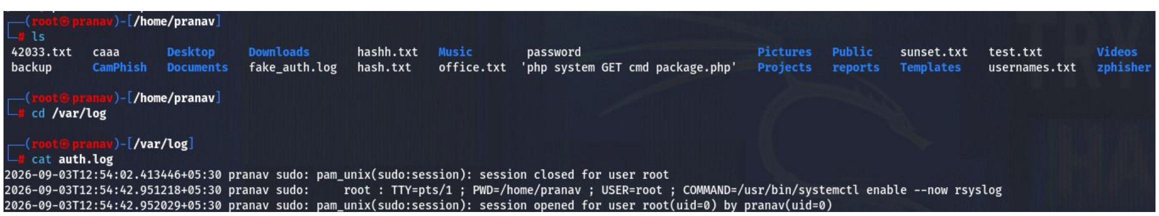
Task 5: Authentication Logs (/var/log/auth.log)

2 login/sudo examples:

```
2026-09-03T12:54:42.951218+05:30 pranav sudo: root : TTY=pts/1 ; PWD=/home/pranav ; USER=root ; COMMAND=/usr/bin/systemctl enable --now rsyslog

2026-09-03T12:54:42.952029+05:30 pranav sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pranav(uid=0)
```

Screenshot: cd /var/log → cat auth.log (excerpt)


```
┌──(root㉿pranav)-[/home/pranav]
└─# ls
42033.txt  caaa      Desktop    Downloads      hashh.txt  Music       password                          Pictures  Public   sunset.txt  test.txt       Videos
backup     CamPhish  Documents  fake_auth.log  hash.txt   office.txt  'php system GET cmd package.php'  Projects  reports  Templates   usernames.txt  zphisher

┌──(root㉿pranav)-[/home/pranav]
└─# cd /var/log

┌──(root㉿pranav)-[/var/log]
└─# cat auth.log
2026-09-03T12:54:02.413446+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T12:54:42.951218+05:30 pranav sudo:     root : TTY=pts/1 ; PWD=/home/pranav ; USER=root ; COMMAND=/usr/bin/systemctl enable --now rsyslog
2026-09-03T12:54:42.952029+05:30 pranav sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pranav(uid=0)
```

Task 6: Keyword Search

Command:

```
grep -i "Failed" /var/log/auth.log
```

Failed attempts:
6 total entries (unix_chkpwd failed checks for user pranav).

Indication:
Indicates user typing errors during authentication or localized password guessing attempts.

Screenshot: grep -i "Failed" /var/log/auth.log output

```
┌──(root㉿pranav)-[/var/log]
└─# grep -i "Failed" /var/log/auth.log
2026-09-03T13:04:43.123007+05:30 pranav unix_chkpwd[104593]: password check failed for user (pranav)
2026-09-03T13:04:47.413543+05:30 pranav unix_chkpwd[104613]: password check failed for user (pranav)
2026-09-03T14:25:03.922607+05:30 pranav unix_chkpwd[4874]: password check failed for user (pranav)
2026-09-03T14:25:08.577683+05:30 pranav unix_chkpwd[4900]: password check failed for user (pranav)
2026-09-03T14:25:13.349054+05:30 pranav unix_chkpwd[4920]: password check failed for user (pranav)
2026-09-03T14:35:10.050317+05:30 pranav unix_chkpwd[10649]: password check failed for user (pranav)
```

LEVEL 3 – ADVANCED (SOC THINKING)

Task 7: Suspicious Activity Detection

Findings:
Rapid privilege escalations (sudo to root via /usr/bin/su) across multiple pseudo-terminal sessions (pts/1, pts/2, pts/3) combined with failed PAM checks.

SOC significance:
Rapid creation of multiple root sessions within seconds can signal privilege escalation exploits, persistent backdoors, or session hijacking.

Task 8: Pattern Identification

User pranav repeatedly attempts to switch to root via su, accompanied by clustered failed authentication timestamps followed by successful root session initialization.

Screenshot: cat /var/log/auth.log (full session showing sudo/su escalation pattern)

```
└─# cat /var/log/auth.log
2026-09-03T12:54:02.413446+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T12:54:42.951218+05:30 pranav sudo:     root : TTY=pts/1 ; PWD=/home/pranav ; USER=root ; COMMAND=/usr/bin/systemctl enable --now rsyslog
2026-09-03T12:54:42.952029+05:30 pranav sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pranav(uid=0)
2026-09-03T12:54:43.210840+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T12:55:01.928411+05:30 pranav CRON[100841]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-09-03T12:55:01.939465+05:30 pranav CRON[100841]: pam_unix(cron:session): session closed for user root
2026-09-03T13:01:08.903250+05:30 pranav sudo:     root : TTY=pts/1 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/whoami
2026-09-03T13:01:08.910165+05:30 pranav sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pranav(uid=0)
2026-09-03T13:01:08.913273+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T13:01:51.053547+05:30 pranav sudo:     root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/whoami
2026-09-03T13:01:51.054375+05:30 pranav sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pranav(uid=0)
2026-09-03T13:01:51.060139+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T13:01:57.644937+05:30 pranav sudo:     root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/su
2026-09-03T13:01:57.645023+05:30 pranav sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pranav(uid=0)
2026-09-03T13:01:57.651552+05:30 pranav su[103153]: (to root) root on pts/2
2026-09-03T13:01:57.651691+05:30 pranav su[103153]: pam_unix(su:session): session opened for user root(uid=0) by pranav(uid=0)
2026-09-03T13:03:08.858229+05:30 pranav sudo:     root : TTY=pts/2 ; PWD=/root ; USER=root ; COMMAND=/usr/bin/su
2026-09-03T13:03:08.858790+05:30 pranav sudo: pam_unix(sudo:session): session opened for user root(uid=0) by pranav(uid=0)
2026-09-03T13:03:08.874865+05:30 pranav su[103572]: (to root) root on pts/3
2026-09-03T13:03:08.875122+05:30 pranav su[103572]: pam_unix(su:session): session opened for user root(uid=0) by pranav(uid=0)
2026-09-03T13:04:28.916034+05:30 pranav su[103572]: pam_unix(su:session): session closed for user root
2026-09-03T13:04:28.917260+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T13:04:28.931883+05:30 pranav su[103153]: pam_unix(su:session): session closed for user root
2026-09-03T13:04:28.933093+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T13:04:28.949219+05:30 pranav su[4154]: pam_unix(su:session): session closed for user root
2026-09-03T13:04:28.959048+05:30 pranav sudo: pam_unix(sudo:session): session closed for user root
2026-09-03T13:04:28.966098+05:30 pranav systemd-logind[592]: Session c2 logged out. Waiting for processes to exit.
2026-09-03T13:04:28.966343+05:30 pranav systemd-logind[592]: Removed session c2.
2026-09-03T13:04:39.323692+05:30 pranav systemd-logind[592]: Removed session 4.
```

Task 9: Fake Log Generation (fake_auth.log)

```
2026-09-03T15:10:01+05:30 server sshd[12401]: Failed password for user admin from 192.168.1.10 port 49152 ssh2
2026-09-03T15:10:03+05:30 server sshd[12405]: Failed password for user admin from 192.168.1.10 port 49154 ssh2
2026-09-03T15:10:05+05:30 server sshd[12409]: Failed password for user admin from 192.168.1.10 port 49156 ssh2
2026-09-03T15:10:08+05:30 server sshd[12412]: Failed password for user root from 192.168.1.10 port 49160 ssh2
2026-09-03T15:10:12+05:30 server sshd[12418]: Accepted password for user root from 192.168.1.10 port 49162 ssh2
```

CRITICAL THINKING & BONUS

Task 10: 50 Failed Attempts in 1 Minute

A spike of 50 failed attempts in 60 seconds indicates an automated brute-force attack or credential stuffing campaign using tools like Hydra or Medusa. It could also stem from a broken service script repeating bad credentials. A SOC analyst must immediately block the source IP, verify account lockouts, and check whether any successful attempt followed the failures.

Task 11: Successful vs. Failed Login

Successful Login	Failed Login
Validates credentials against system PAM, generates session opened / Accepted password entries, and grants user access.	Rejects non-matching credentials, outputs Failed password or password check Failed, and denies system access.

Task 12: Cybersecurity Importance of Logs

- Provide real-time operational visibility into systems and services.
- Form the core foundation for incident response and digital forensics.
- Fulfill regulatory compliance requirements (e.g., ISO 27001, PCI-DSS).